

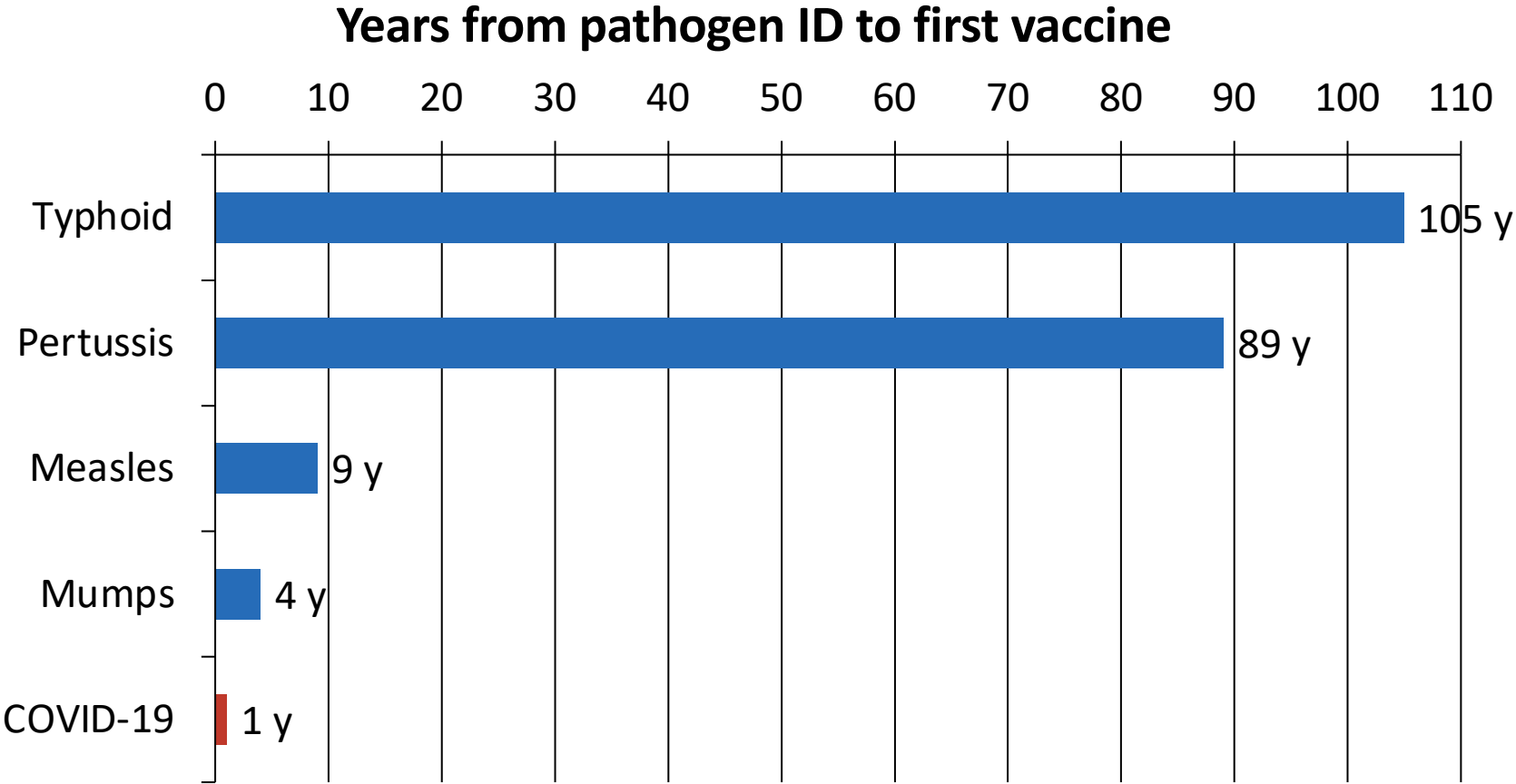
The Power of Vaccination: From Smallpox Eradication to COVID-19 in Recor

1796–2020 | A data-driven overview for public health professionals and policy analysts

Vaccination transformed global health in two centuries:

- Development speed accelerated from 105 years to under 1 year
- Multiple deadly diseases reached 100% reduction in US cases/deaths
- Herd immunity protects vulnerable populations
- Coverage and access gaps remain for key preventable diseases

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating



COVID-19
< 1 year
Record licensure timeline

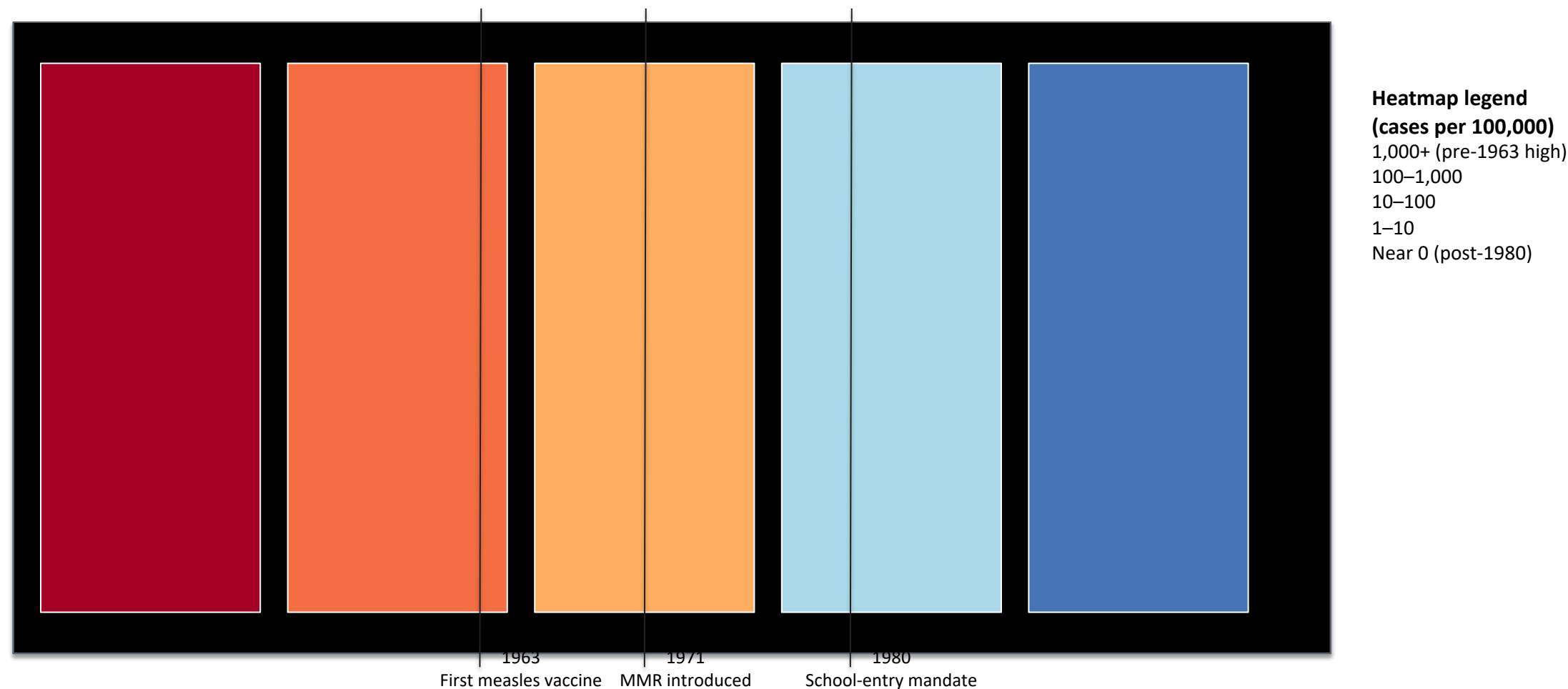
Source: Our World in Data (as provided in source brief)

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the

Disease	Pre-vaccine annual cases	Current annual cases	Case reduction	Pre-vaccine annual deaths	Death reduction
Diphtheria	21,053	0	100%	1,822	100%
Polio (paralytic)	16,316	0	100%	1,879	100%
Smallpox	48,164	0	100%	152	100%
Pneumococcal disease	63,067	37,999	40.5%	6,372	31.3%
Hepatitis A	117,333	12,474	89.0%	117	(not specified)
Acute hepatitis B	66,232	10,354	83.9%	237	(not specified)

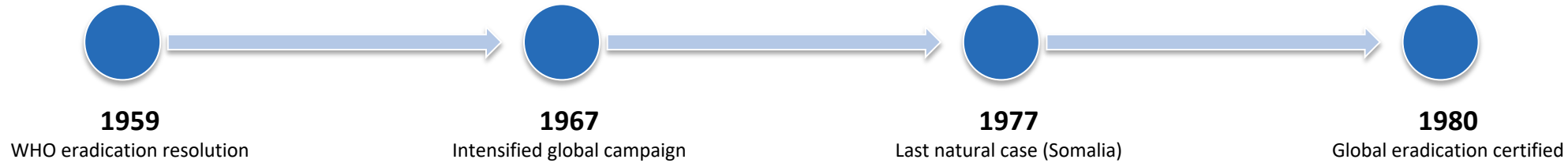
Source: Our World in Data (as provided in source brief)

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates



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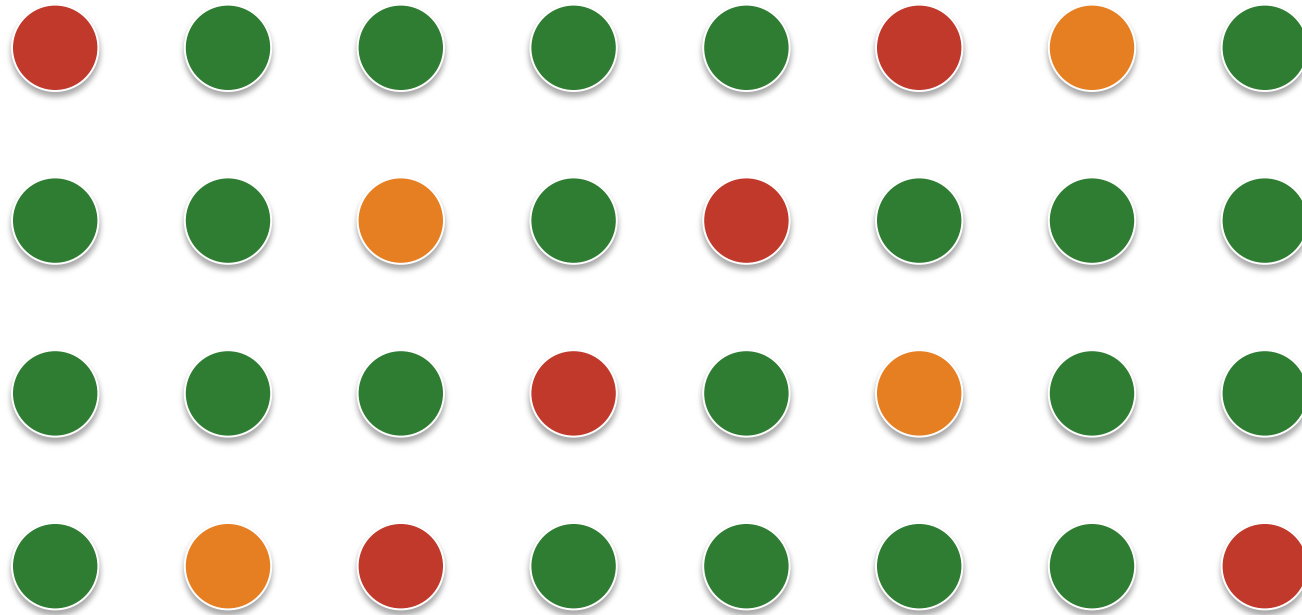
Smallpox: The Only Human Disease Eradicated Through Vaccination



Why eradication was possible:

- Human-only reservoir
- Distinctive, visible symptoms
- Effective, low-cost vaccine

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

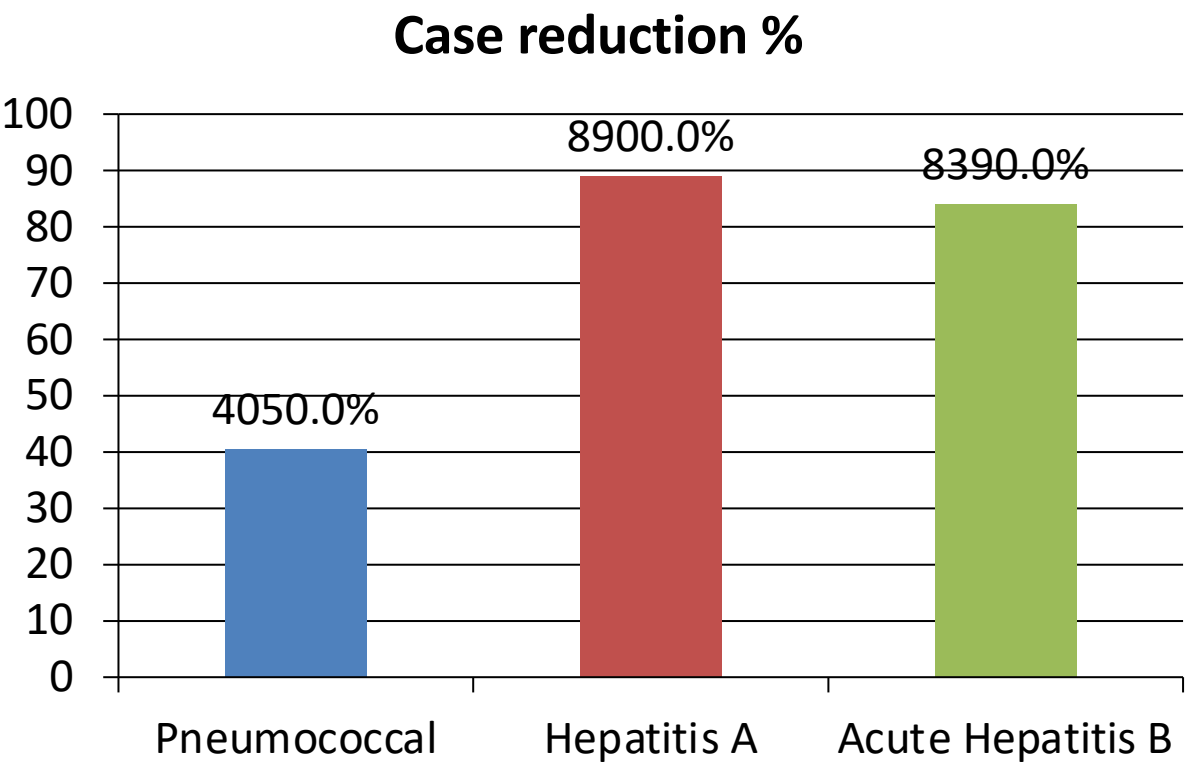


Mechanism

- High vaccination coverage breaks transmission chains
- Unvaccinated vulnerable people are indirectly protected
- Critical for newborns and immunocompromised patients
- Protects patients on chemotherapy

Green: vaccinated | Red: susceptible | Orange: vulnerable (cannot be vaccinated)

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction



Persistent barriers

- Uneven global access
- Supply shortages
- Coverage gaps in high-risk groups
- Low trust from poor scientific understanding

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job

- Vaccines achieved 100% elimination for diphtheria, smallpox, and polio in the US.
- Development timelines collapsed from 105 years (typhoid) to under 1 year (COVID-19).
- Measles incidence fell from high endemic levels to near-zero after mandates.
- Herd immunity protects newborns and immunocompromised populations.
- Coverage gaps remain for pneumococcal disease, hepatitis A, and hepatitis B.

Call to action: Strengthen access, sustain coverage, and build trust to maximize vaccine impact globally.